SQL PROJECT

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OVERVIEW

You are working as a Data Analyst in a growing E-commerce company in Egypt.

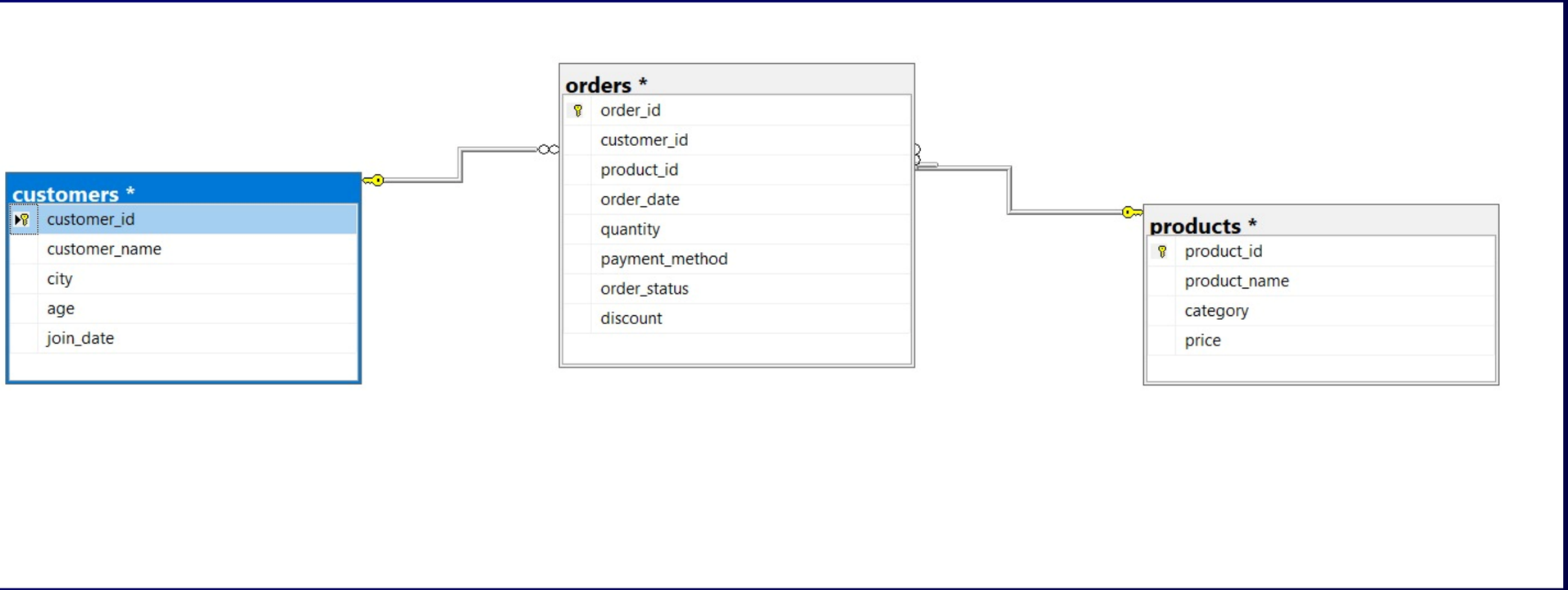
The company sells a variety of products such as electronics, accessories, and furniture. Customers from different cities place orders using multiple payment methods like Cash, Visa, and Instapay

Each order may include discounts, and orders can have different statuses such as Delivered, Cancelled, or Pending

The management team wants to better understand customer behavior, sales performance, and business trends in order to make data-driven decisions.

Your task is to analyze the data and answer the following business questions using SQL.

DATA ARCHITECTURE



BUSINESS QUESTIONS

Q1. Retrieve all orders that occurred in 2023.

```
select *
from orders
where year (order_date) = 2023
```

order_id	customer_id	product_id	order_date	quantity	payment_method	order_status	discount
1	81	10	2023-10-25	3	Instapay	Delivered	0
2	90	7	2023-02-05	1	Cash	Delivered	5
3	8	1	2023-03-25	3	Cash	Delivered	0
4	93	7	2023-09-20	2	Cash	Delivered	10
5	26	10	2023-11-06	4	Cash	Pending	15
6	74	5	2023-06-21	1	Cash	Delivered	0
7	90	6	2023-06-22	2	Visa	Delivered	15
8	34	10	2023-12-28	4	Cash	Delivered	0
9	7	7	2023-02-16	1	Instapay	Delivered	5
10	68	14	2023-05-21	3	Instapay	Delivered	15
11	58	15	2023-04-28	2	Visa	Delivered	0
12	75	5	2023-10-04	4	Instapay	Delivered	20
13	29	7	2023-08-02	1	Visa	Delivered	0
14	36	9	2023-08-08	1	Visa	Delivered	0
15	89	3	2023-01-26	3	Cash	Delivered	15
16	21	8	2023-01-09	4	Cash	Delivered	0
17	36	2	2023-09-28	3	Visa	Cancelled	20
18	10	11	2023-01-10	3	Visa	Delivered	20
19	73	7	2023-05-01	4	Cash	Delivered	0
20	24	13	2023-05-05	4	Visa	Delivered	5
21	64	2	2023-06-19	4	Visa	Delivered	0
22	99	15	2023-03-13	2	Cash	Pending	0
23	49	14	2023-06-01	4	Instapay	Delivered	0
24	99	4	2023-07-27	1	Visa	Delivered	0
25	36	7	2023-02-18	3	Cash	Delivered	0

BUSINESS QUESTIONS

Q2. Retrieve all orders with status = 'Delivered'.

```
select *
from orders
where order_status = 'Delivered'
```

order_id	customer_id	product_id	order_date	quantity	payment_method	order_status	discount
1	81	10	2023-10-25	3	Instapay	Delivered	0
2	90	7	2023-02-05	1	Cash	Delivered	5
3	8	1	2023-03-25	3	Cash	Delivered	0
4	93	7	2023-09-20	2	Cash	Delivered	10
6	74	5	2023-06-21	1	Cash	Delivered	0
7	90	6	2023-06-22	2	Visa	Delivered	15
8	34	10	2023-12-28	4	Cash	Delivered	0
9	7	7	2023-02-16	1	Instapay	Delivered	5
10	68	14	2023-05-21	3	Instapay	Delivered	15
11	58	15	2023-04-28	2	Visa	Delivered	0
12	75	5	2023-10-04	4	Instapay	Delivered	20
13	29	7	2023-08-02	1	Visa	Delivered	0
14	36	9	2023-08-08	1	Visa	Delivered	0
15	89	3	2023-01-26	3	Cash	Delivered	15
16	21	8	2023-01-09	4	Cash	Delivered	0
18	10	11	2023-01-10	3	Visa	Delivered	20
19	73	7	2023-05-01	4	Cash	Delivered	0
20	24	13	2023-05-05	4	Visa	Delivered	5
21	64	2	2023-06-19	4	Visa	Delivered	0
23	49	14	2023-06-01	4	Instapay	Delivered	0

BUSINESS QUESTIONS

Q3. Retrieve all customers who live in Cairo.

```
select *
from customers
where city = 'Cairo'
```

customer_id	customer_name	city	age	join_date
15	Mona Adel	Cairo	52	2023-06-16
16	Dina Mostafa	Cairo	58	2021-09-29
19	Ibrahim Adel	Cairo	46	2022-03-23
22	Ibrahim Samir	Cairo	50	2023-05-11
23	Mohamed Ibrahim	Cairo	38	2021-09-04
33	Mona Ali	Cairo	57	2021-02-16
36	Yara Adel	Cairo	39	2022-03-29
38	Aya Ali	Cairo	19	2021-08-06
45	Heba Fathy	Cairo	50	2021-09-11
46	Mahmoud Mahmoud	Cairo	26	2023-01-04
52	Ahmed Adel	Cairo	42	2023-04-17
55	Mahmoud Hassan	Cairo	24	2021-06-29
62	Ali Hassan	Cairo	29	2021-11-21
64	Omar Adel	Cairo	32	2021-01-16
73	Ahmed Mostafa	Cairo	39	2023-05-02
75	Mahmoud Adel	Cairo	23	2022-11-21
77	Ibrahim Saeed	Cairo	54	2022-01-15
83	Dina Adel	Cairo	23	2023-04-15
84	Mohamed Mahmoud	Cairo	49	2022-07-31

BUSINESS QUESTIONS

Q4.Retrieve all products with price greater than 5000.

```
select *
from Products
where price > 5000
```

product_id	product_name	category	price
1	Laptop	Electronics	20000
2	iPhone	Electronics	30000
3	Samsung Phone	Electronics	15000
7	Monitor	Electronics	6000
8	Tablet	Electronics	10000
10	Camera	Electronics	12000

BUSINESS QUESTIONS

Q5. Retrieve all orders where the discount is greater than 10%.

```
select *
from orders
where discount > 10
```

order_id	customer_id	product_id	order_date	quantity	payment_method	order_status	discount
5	26	10	2023-11-06	4	Cash	Pending	15
7	90	6	2023-06-22	2	Visa	Delivered	15
10	68	14	2023-05-21	3	Instapay	Delivered	15
12	75	5	2023-10-04	4	Instapay	Delivered	20
15	89	3	2023-01-26	3	Cash	Delivered	15
17	36	2	2023-09-28	3	Visa	Cancelled	20
18	10	11	2023-01-10	3	Visa	Delivered	20
27	96	2	2023-12-13	4	Cash	Delivered	20
34	55	8	2023-03-01	1	Cash	Delivered	20
38	99	13	2023-05-26	2	Cash	Cancelled	15
45	13	7	2023-08-14	2	Instapay	Delivered	15
59	52	13	2023-11-16	2	Visa	Delivered	20
62	69	14	2023-03-26	1	Visa	Delivered	15
63	99	2	2023-05-26	4	Cash	Delivered	20
79	95	13	2023-04-19	3	Cash	Delivered	20
82	87	13	2023-08-17	1	Cash	Pending	20
85	18	12	2023-03-14	2	Cash	Delivered	15
102	37	13	2023-03-08	1	Cash	Delivered	15
106	65	2	2023-12-08	3	Cash	Delivered	15
113	36	5	2023-07-28	1	Cash	Delivered	20
116	19	9	2023-10-13	4	Visa	Delivered	15
131	92	7	2023-01-12	4	Visa	Delivered	15

BUSINESS QUESTIONS

Q6. Count the total number of orders.

```
select count(order_id) As Total_orders
from orders
```

Total_orders
1000

BUSINESS QUESTIONS

Q7. Calculate the total quantity of products sold.

```
select sum(quantity) As Total_Quantity
from orders
```

Total_Quantity
2478

BUSINESS QUESTIONS

Q8. Calculate the average product price.

```
select category , Avg(price) As Avg_price
from products
group by category
```

category	Avg_price
Electronics	12687.5000
Accessories	1000.0000
Furniture	1700.0000

BUSINESS QUESTIONS

Q9. Count the number of orders for each order status (Delivered, Cancelled, Pending).

```
select order_status , count(order_id) AS order_count
from orders
group by order_status
```

order_status	order_count
Delivered	815
Pending	86
Cancelled	99

BUSINESS QUESTIONS

Q10. Count the number of customers in each city.

```
select city,count(customer_id) AS count_customer
from customers
group by city
```

city	count_customer
Tanta	7
Alexandria	21
Mansoura	18
Giza	10
Ismailia	9
Cairo	19
Zagazig	16

BUSINESS QUESTIONS

Q11. Retrieve each order along with the customer name who made it.

```
select  O.order_id , C.customer_name
from orders O
join customers C
on C.customer_id = O.customer_id
```

order_id	customer_name
1	Ibrahim Ibrahim
2	Ahmed Mahmoud
3	Mohamed Fathy
4	Youssef Adel
5	Fatma Mahmoud
6	Dina Mostafa
7	Ahmed Mahmoud
8	Tarek Ibrahim
9	Mahmoud Khaled
10	Mohamed Samir
11	Ahmed Saeed
12	Mahmoud Adel
13	Hassan Ibrahim
14	Yara Adel
15	Omar Hassan
16	Yara Ali
17	Yara Adel
18	Mohamed Saeed
19	Ahmed Mostafa
20	Fatma Adel
21	Omar Adel
22	Sara Hassan

BUSINESS QUESTIONS

Q12. Retrieve each product along with the total quantity sold.

```
select P.product_name , sum(O.quantity) AS Total_quantity
from products P
join orders O
on O.product_id = P.product_id
group by product_name
```

product_name	Total_quantity
Camera	169
Monitor	152
Laptop	172
Keyboard	152
Mouse	143
Office Chair	174
Desk Lamp	169
Charger	156
Samsung Phone	152
Tablet	147
iPhone	219
Smart Watch	168
Printer	184
Headphones	131
Speaker	190

BUSINESS QUESTIONS

Q13. Calculate total sales for each product (price × quantity × (1 - discount/100)).

```
select P.product_name , sum(P.price * O.quantity * (1 - O.discount/100)) AS Total_Sales
from  orders O
join products P
on O.Product_id = P.Product_id
group by P.product_name
```

product_name	Total_Sales
Camera	1934400.0000
Monitor	878700.0000
Laptop	3313000.0000
Keyboard	143800.0000
Mouse	54680.0000
Office Chair	410500.0000
Desk Lamp	143910.0000
Charger	44385.0000
Samsung Phone	2181000.0000
Tablet	1407000.0000
iPhone	6141000.0000
Smart Watch	795750.0000
Printer	603750.0000
Headphones	101920.0000
Speaker	454500.0000

BUSINESS QUESTIONS

Q14. Find the top 5 customers based on total spending..

```
select C.customer_name , sum(P.price * O.quantity * (1 - O.discount/100)) AS Total_Spending
from products P
join orders O
on P.product_id = O.product_id
join customers C
on O.customer_id = C.customer_id
group by customer_name
order by Total_Spending desc
limit 5
```

customer_name	Total_Spending
Mona Ali	639520.0000
Sara Fathy	593075.0000
Mona Fathy	538605.0000
Dina Mostafa	530935.0000
Yara Adel	528010.0000

BUSINESS QUESTIONS

Q15. Retrieve customers who made more than 3 orders

```
select customer_id , count(order_id) AS Order_count
from orders
group by customer_id
having Order_count > 3
```

customer_id	Order_count
81	10
90	13
8	13
93	6
26	19
74	15
34	11
7	8
68	7
58	7
75	14
29	15
36	16
89	11
21	10
10	9
73	10

BUSINESS QUESTIONS

Q16. Find the total sales for each city.

```
select C.city , sum(P.price * O.quantity * (1 - O.discount/100)) AS Total_Sales
from orders O
join Customers C
on O.customer_id = c.customer_id
join products P
on O.product_id = P.product_id
group by C.city
order by C.city
```

city	Total_Sales
Alexandria	4394550.0000
Cairo	3477355.0000
Giza	1367110.0000
Ismailia	1843385.0000
Mansoura	3098865.0000
Tanta	1126805.0000
Zagazig	3300225.0000

BUSINESS QUESTIONS

Q17. Find the most sold product (by quantity).

```
select product_id , sum(quantity) As Total_Quantity
from orders
group by product_id
order by  Total_Quantity desc
limit 1
```

product_id	Total_Quantity
2	219

BUSINESS QUESTIONS

Q18. Calculate the total revenue generated for each payment method.

```
select O.payment_method , sum(P.price * O.quantity * (1 - O.discount/100)) AS Total_Revenue
from orders O
join products P
on O.product_id = P.product_id
group by O.payment_method
order by O.Payment_method
```

payment_method	Total_Revenue
Cash	8771795.0000
Instapay	2070725.0000
Visa	7765775.0000

BUSINESS QUESTIONS

Q19. Find the monthly total sales trend

```
select
year(order_date) As Year ,
Month(order_date) As Month ,
sum(P.price * O.quantity * (1 - O.discount/100)) AS Total_Sales
from orders O
join products P
on O.product_id = P.product_id
group by Month , year
order by Month , year
```

Year	Month	Total_Sales
2023	1	1528390.0000
2023	2	1456680.0000
2023	3	1907400.0000
2023	4	1504665.0000
2023	5	1708405.0000
2023	6	2087070.0000
2023	7	1253230.0000
2023	8	1271915.0000
2023	9	1969570.0000
2023	10	1214510.0000
2023	11	1374110.0000
2023	12	1332350.0000

BUSINESS QUESTIONS

Q20. Identify high-value customers who: made more than 3 orders and spent more than 20000

```
select O.customer_id , count(O.order_id) AS Order_count ,sum(P.price * O.quantity * (1 - O.discount/100)) AS Total_Sales
from orders O
join products P
on O.product_id = P.product_id
group by customer_id
having count(O.order_id) > 3 and sum(P.price * O.quantity * (1 - O.discount/100)) > 20000
```

customer_id	Order_count	Total_Sales
81	10	153070.0000
90	13	248005.0000
8	13	157315.0000
93	6	131655.0000
26	19	312455.0000
74	15	366515.0000
34	11	273450.0000
7	8	240050.0000
68	7	40900.0000
58	7	164030.0000
75	14	256000.0000
29	15	453350.0000
36	16	428650.0000
89	11	273345.0000
21	10	118590.0000
10	9	134585.0000
73	10	93530.0000
24	9	220850.0000
64	10	265045.0000
99	12	341700.0000
49	13	202900.0000
82	9	224025.0000
96	8	283735.0000
23	12	229845.0000
62	14	273925.0000

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THANK YOU